The two remaining domain pathologies in properly infinite von Neumann algebras

Autonomous proof completion for arXiv:2311.16170

11 August 2026

Abstract

Ghosh and Nayak ask three domain-pathology questions for affiliated operators of a properly infinite von Neumann algebra, answer the third, and leave the first two to the reader. We give the short complete argument. Their normal embedding of $\mathcal{B}(\ell^2)$ and canonical affiliated-operator functor transport von Neumann's classical disjoint-domain pair and Chernoff's square-trivial symmetric operator while preserving products, domains, lattice intersections, closedness, and the relevant order and adjoint properties.

1 Statement

Let $\mathscr{M} \subseteq \mathcal{B}(\mathcal{H})$ be a properly infinite von Neumann algebra and let $\mathscr{M}_{\text{aff}}^c$ denote its closed affiliated operators in the notation of the source paper.

Theorem 1. *Both Questions 1 and 2 of Section 8 of the source have affirmative answers:*

1. *There are a positive self-adjoint $T \in \mathscr{M}_{\text{aff}}^c$ and a unitary $U \in \mathscr{M}$ such that*
$$\operatorname{dom}(T) \cap \operatorname{dom}(UTU^*) = \{0\}_{\mathcal{H}}.$$
2. *There is a densely-defined closed symmetric $A \in \mathscr{M}_{\text{aff}}^c$ such that*
$$\operatorname{dom}(A^2) = \{0\}_{\mathcal{H}}.$$

2 The transport identities

By Proposition 8.1 of the source there is a unital normal embedding

$$\Phi : \mathcal{B}(\ell^2(\mathbb{N})) \longrightarrow \mathscr{M}.$$

Write Φ_{aff} for its canonical extension to affiliated operators and Φ_{aff}^s for its extension to affiliated operator ranges. The source proves the following facts:

$$\operatorname{dom}(\Phi_{\text{aff}}(S)) = \Phi_{\text{aff}}^s(\operatorname{dom} S), \tag{1}$$
$$\Phi_{\text{aff}}(S_1 S_2) = \Phi_{\text{aff}}(S_1)\Phi_{\text{aff}}(S_2), \tag{2}$$
$$\Phi_{\text{aff}}(S^*) = \Phi_{\text{aff}}(S)^*. \tag{3}$$

Here (1) is Theorem 5.5(i), and (2)–(3) are Theorem 4.14(ii),(iv). Theorem 3.10 says that Φ_{aff}^s preserves intersections and

$$\Phi_{\text{aff}}^s(\{0\}_{\ell^2}) = \{0\}_{\mathcal{H}}. \tag{4}$$

Theorem 5.5 preserves dense definition and closedness; Theorem 6.6 preserves symmetry, positivity, and self-adjointness.

3 Question 1

The classical theorem of von Neumann cited by the source supplies an unbounded positive self-adjoint operator T_0 and a unitary U_0 on $\ell^2(\mathbb{N})$ such that

$$\operatorname{dom}(T_0) \cap \operatorname{dom}(U_0 T_0 U_0^*) = \{0\}_{\ell^2}. \tag{5}$$

(This is the reason the source calls Question 1 a weaker version of von Neumann's result.) Put

$$T = \Phi_{\mathrm{aff}}(T_0), \qquad U = \Phi(U_0).$$

Because Φ is a unital $*$-homomorphism, U is unitary. The preservation results above show that T is positive, self-adjoint, closed, and affiliated, and repeated use of (2)–(3) gives

$$UTU^* = \Phi_{\mathrm{aff}}(U_0 T_0 U_0^*).$$

Consequently, by (1), lattice preservation, (5), and (4),

$$\begin{aligned} \operatorname{dom}(T) \cap \operatorname{dom}(UTU^*) &= \Phi^s_{\mathrm{aff}}(\operatorname{dom} T_0) \cap \Phi^s_{\mathrm{aff}}(\operatorname{dom}(U_0 T_0 U_0^*)) \\ &= \Phi^s_{\mathrm{aff}}(\operatorname{dom} T_0 \cap \operatorname{dom}(U_0 T_0 U_0^*)) = \{0\}_{\mathcal{H}}. \end{aligned}$$

4 Question 2

Chernoff's explicit example cited by the source is a densely-defined, semibounded, closed symmetric operator A_0 on $\ell^2(\mathbb{N})$ such that

$$\operatorname{dom}(A_0^2) = \{0\}_{\ell^2}. \tag{6}$$

Set $A = \Phi_{\mathrm{aff}}(A_0)$. Theorems 5.5 and 6.6 show that A is densely-defined, closed, symmetric, and affiliated. By exact product preservation,

$$A^2 = \Phi_{\mathrm{aff}}(A_0^2).$$

Using (1), (6), and (4) now gives

$$\operatorname{dom}(A^2) = \Phi^s_{\mathrm{aff}}(\operatorname{dom}(A_0^2)) = \Phi^s_{\mathrm{aff}}(\{0\}_{\ell^2}) = \{0\}_{\mathcal{H}}.$$

This proves Theorem 1. $\square$

Scope and references

The exact questions occur on source PDF pages 36–37; the source explicitly leaves them to the reader on page 37. The Hilbert-space inputs are J. von Neumann, *Zur Theorie der unbeschränkten Matrizen*, J. Reine Angew. Math. 161 (1929), 208–236, and P. R. Chernoff, *A semibounded closed symmetric operator whose square has trivial domain*, Proc. Amer. Math. Soc. 89 (1983), 289–290. The proof is a completion of the indicated exercise, not a claim that those classical pathologies are new.